

Herpesvirus Infections - Factoid Sheet

Week 8 • Core Lecture 4 • current Newman-Whitlow deck controls

Family rule: enveloped dsDNA + lifelong latency/reactivation. Alpha = HSV1/2/VZV • Beta = CMV/HHV6/7 • Gamma = EBV/HHV8.

Latency & Core Identity

Virus	Classic latency site / deck anchor	Signature disease pattern
HSV-1	Trigeminal ganglion in deck	Oral disease, keratitis, temporal-lobe encephalitis
HSV-2	Sacral ganglia - handout enrichment	Genital herpes, neonatal disease
VZV	Sensory/dorsal-root ganglia	Primary varicella → reactivation zoster
EBV	B cells - handout enrichment	Heterophile-positive infectious mononucleosis
CMV	Mononuclear cells - handout enrichment	Heterophile-negative mono; congenital/opportunistic disease

HSV: Must-Recognize

Pattern	High-yield clue
HSV-1 primary	Childhood saliva; gingivostomatitis; painful oral lesions; latency usually trigeminal
HSV-1 recurrence	Cold sores after fever/stress/sun/trauma/immunosuppression
HSV-2	Painful grouped genital vesicles → pustules/ulcers; asymptomatic shedding; recurrences generally milder
HSV encephalitis	Temporal lobe + fever/behavior/seizures + mixed CSF pleocytosis + ↑protein + RBCs → CSF HSV PCR → IV acyclovir
Other	Whitlow • keratitis/dendritic ulcer • neonatal herpes • eczema herpeticum

VZV: Chickenpox vs Shingles

Primary varicella	Reactivation zoster
Respiratory/contact; 10-14 d incubation; rash in crops; lesions in different stages; contagious until all crusted	Pain/dysesthesia → dermatomal vesicular rash; thoracic/lumbar common; disseminates in immunocompromised
VARIVAX prevention; VZIG for high-risk exposures	Famciclovir/valacyclovir; IV acyclovir if immunocompromised; PHN = most common late complication

Shingrix: recombinant subunit + adjuvant, 2-dose series. Lecture recommendation: healthy adults >50 and immunocompromised adults >18.

EBV vs CMV Mononucleosis

Feature	EBV	CMV
Heterophile / Monospot	Positive	Negative
Atypical lymphocytes	Yes - reactive CD8 T cells	Yes
Clinical anchor	Fever, pharyngitis, adenopathy, splenomegaly; aminopenicillin rash	Mono-like syndrome; less specific throat syndrome in deck
Management	Rest; no contact sports; no beneficial specific antiviral per deck	No specific Rx in healthy host

CMV + Other HHVs + Treatment + Mpox

CMV

Setting	Lecture clue
Congenital	Jaundice, hepatosplenomegaly, rash, microcephaly, cerebral calcification, chorioretinitis, seizures. Handout adds SN hearing loss + periventricular calcifications.
Immunosuppressed	HIV/BMT/organ transplant/chemo; retinitis, pneumonitis, encephalitis, esophagitis, colitis
Diagnosis	Quantitative CMV PCR, culture/serology, biopsy; owl's-eye intranuclear inclusion
Treatment list in deck	Ganciclovir/valganciclovir • foscarnet • cidofovir • CMV Ig • letermovir prevention in BMT • maribavir resistant CMV • no vaccine

HHV-6/7/8

Virus	Classic clue
HHV-6/7	Roseola: high fever → defervescence → maculopapular rash ; febrile seizures/encephalitis possible; HHV-7 similar
HHV-8	Kaposi sarcoma • primary effusion lymphoma • multicentric Castleman disease

Antiviral / Prevention Map

Clinical use	Current-deck therapy	Handout-only board enrichment
HSV / VZV	Acyclovir, valacyclovir, famciclovir	Viral thymidine kinase activation → DNA polymerase inhibition/chain termination
CMV	Ganciclovir/valganciclovir; resistant disease options above	UL97 activation; marrow suppression is classic ganciclovir toxicity
Resistant herpesvirus	Foscarnet/cidofovir listed	Do not require viral kinase; nephrotoxicity is classic
Vaccines	VARIVAX for varicella; Shingrix for zoster	Shingrix = recombinant subunit; old Zostavax was live

Diagnosis + CSF

Test	Use
PCR	Blood, CSF, other fluids; test of choice in HSV encephalitis vignette
Serology	IgM acute; IgG prior exposure; EBV heterophile/Monospot in lecture
Tzanck	Multinucleated giant cells; supports HSV/VZV but does not speciate
Histology	CMV owl's-eye; tissue biopsy for organ disease

Mpox

Deck phenotype

Orthopoxvirus, not herpes. Direct skin-to-skin contact with lesions; often genital in U.S. cases; painful vesicles at inoculation site → crust.

Dx / prevention

Lesion PCR. JYNNEOS = non-replicating modified vaccinia vaccine, 2 doses, for exposed or at-risk people.

Source conflict: older MCG 2029 handout says tecovirimat may be used in severe Mpox. Current 2026 deck says no known therapy and tecovirimat failed in Mpox trials. **Follow the current deck.**

Final chain: HSV = temporal lobe/RBC CSF • VZV = crops → dermatomal zoster • EBV = heterophile+ • CMV = heterophile-/owl's eye • HHV6/7 = fever then rash • HHV8 = Kaposi • Mpox = lesion PCR + JYNNEOS.

Primary: current 66-slide Newman-Whitlow deck. Official Week 8 LO sheet supplies 3 ScholarRx bricks. MCG 2029 AI Notes “Lecture 7: Herpesvirus Infections” is secondary enrichment only.